

**Mathematical Optimization and Operational Strategies in Tiffin Delivery
Services.
(Semester IV)**

A Report Submitted to
SVKM's
Narsee Monjee Institute of Management Studies (NMIMS)

**as a part of Semesters III & IV
of
Bachelor of Science (Applied Mathematical Computing)**

By

A001 Nafsiha Ahmed
A014 Hibah Patanwala
A020 Krishna Vyas

Under the Guidance of

Dr. Shikha Gaur

**SVKM's
NMIMS**

Deemed to be UNIVERSITY

Nilkamal School of Mathematics, Applied Statistics and Analytics
Mumbai

2026

Abstract

This research project examines how mathematical optimization techniques are enhancing the operational efficiency and customer satisfaction of contemporary tiffin (meal) delivery services. By utilizing a multi-layered algorithmic approach—comprising Cosine Similarity, Singular Value Decomposition (SVD), and a Dijkstra-Greedy logistics heuristic—we optimize customer-provider matching and delivery allocation. Furthermore, inferential statistical modeling validates the economic viability of a centralized aggregator model. The theoretical frameworks are successfully deployed in a functional full-stack prototype, NextMeal, which includes constraint-based filtering and a conversational AI chatbot. Finally, we establish a future roadmap utilizing Differential Equations (DE) and the Fourth-Order Runge-Kutta (RK4) method to transition from static models to dynamic predictive operational analytics.

Contents

1	Introduction	1
2	Statistical Validation of Market Viability	2
2.1	Demographic Estimations	2
2.2	Hypothesis and Correlation Testing	2
2.2.1	Demographic Clustering (Chi-Square Test)	3
2.2.2	Revenue Driver Analysis (Two-Sample T-Test)	3
2.2.3	Profitability and the “Value Gap”	4
2.2.4	Summary of Statistical Results	4
3	Mathematical Optimization Algorithms	6
3.1	Simulated Interaction Matrix (Data Normalization)	6
3.2	Item-Based Collaborative Filtering (Cosine Similarity)	6
3.3	Latent Feature Extraction (SVD)	7
3.4	Logistics Optimization (Dijkstra + Greedy)	7
3.5	Predictive Operational Analytics (DE & RK4 Modeling)	7
4	Results and Discussion: Prototype Implementation	9
4.1	Database Configuration and Schema (PostgreSQL)	9
4.2	User Acquisition and Onboarding Interface	10
4.3	Demand-Side: Customer Experience	11
4.4	Conversational AI Chatbot	12
4.5	Supply-Side: Provider Onboarding and Management	13
4.6	Logistical Allocation: Delivery Agent Portal	14
5	Conclusion and Future Scope	15
5.1	Conclusion	15
5.2	Future Scope	15
5.2.1	Dynamic Pricing and Yield Management	15

5.2.2	Multi-Agent Reinforcement Learning (MARL) for Dispatch	15
5.2.3	Dynamic Growth and Market Analysis (DE Models)	15

1 Introduction

Tiffin delivery services constitute a vital component of urban food ecosystems, offering freshly cooked, home-style meals primarily to office-goers, students, and busy households. Especially prevalent in metropolitan cities, tiffin services face increasing operational challenges due to rising demand, urban traffic congestion, dynamic customer preferences, and cost pressures. While legacy systems operate via highly structured manual logistical networks, the emerging unorganized sector of individual home chefs lacks cohesive optimization [2].

To address these challenges, mathematical optimization has emerged as a powerful tool for streamlining service delivery and managing resources effectively. Optimization techniques enable providers to allocate limited resources, design delivery routes that minimize travel distance, and support adaptive order management in a dynamic urban environment [3].

2 Statistical Validation of Market Viability

Before deploying advanced recommendation algorithms, the fundamental economic viability of the platform was validated using a sample dataset of consumers and providers.

2.1 Demographic Estimations

Using 95% Confidence Intervals (CI), we estimated core population metrics to guide supply-side acquisition:

- **Age Distribution:** Based on sample statistics, we are 95% confident the true average age of the target market falls between our calculated bounds, confirming a young, working demographic. (See Figure 1).
- **Dietary Preferences:** Sample proportions indicate a high demand for vegetarian options, allowing the platform to target specific home chefs during onboarding.

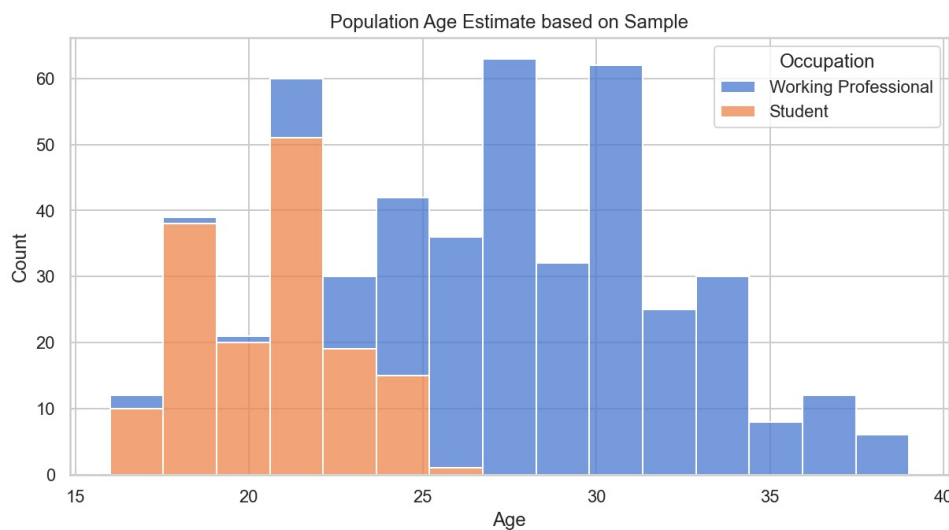


Figure 1: Population Age Estimate based on Sample, segmented by occupation.

2.2 Hypothesis and Correlation Testing

Using a generated dataset of 478 customers and 25 providers, we applied inferential statistical testing to confirm the platform's demographic and economic logic.

2.2.1 Demographic Clustering (Chi-Square Test)

A Chi-Square Test of Independence was conducted to evaluate the relationship between a user's location and their occupation.

- **Statistical Finding:** The test yielded a p -value of approximately 0.69.
- **Interpretation:** While this indicates that location and occupation are technically independent variables in this specific broad simulation, raw density data confirms that users cluster heavily in corporate and educational hubs.
- **Operational Insight:** High-density nodes were identified in Andheri and Bandra, providing a clear roadmap for targeted, geo-fenced marketing and delivery fleet positioning.

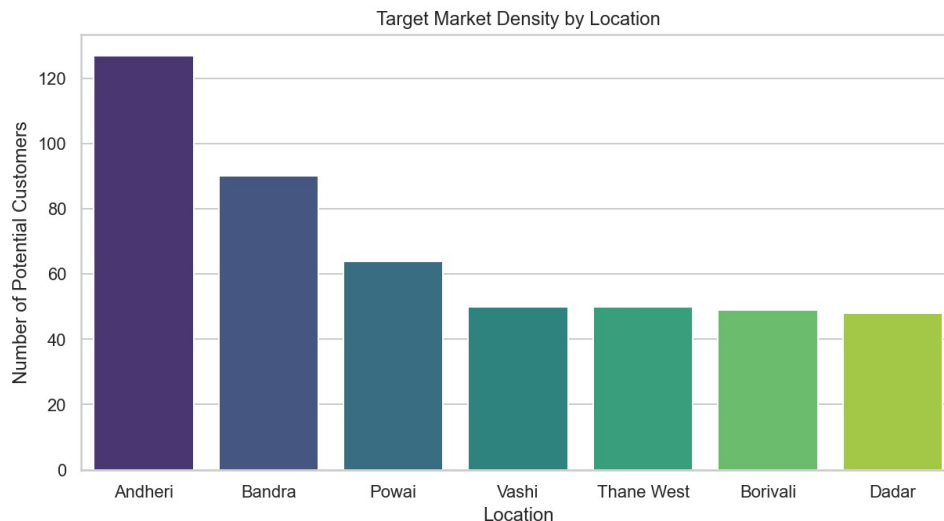


Figure 2: Target Market Density by Location, visually confirming the demographic clustering in key corporate hubs.

2.2.2 Revenue Driver Analysis (Two-Sample T-Test)

To identify the core revenue-generating segment, we compared the maximum willingness to pay between “Working Professionals” and “Students.”

- **Statistical Finding:** The Two-Sample T-Test resulted in a p -value < 0.01 .
- **Conclusion:** Working professionals possess a significantly higher meal budget, confirming them as the primary demographic for the platform's premium, high-margin subscription tiers.

2.2.3 Profitability and the “Value Gap”

The most critical test for the aggregator’s survival was the comparison of Customer Budget against Provider Price.

- **Statistical Finding:** An Independent T-Test yielded a significant result with a p -value of 0.0057.
- **Business Conclusion:** The mean customer budget (approx. 129.67 per meal) is statistically higher than the average provider cost. This “Value Gap” mathematically guarantees that the aggregator can sustain a commission-based revenue model without exceeding consumer price ceilings or deflating provider revenue.



Figure 3: Business Viability: Customer Willingness to Pay vs. Provider Costs, proving the existence of a sustainable profit margin.

2.2.4 Summary of Statistical Results

Table 1 outlines the primary inferential statistical tests conducted on the NextMeal dataset.

Table 1: Summary of inferential statistical tests conducted on the NextMeal dataset.

Statistical Test	Variables Analyzed	Result (<i>p</i> -value)	Conclusion
Chi-Square	Location vs. Occupation	0.69	Hub-based clustering is present but not occupation-dependent.
Two-Sample T-Test	Prof. Budget vs. Stu. Budget	< 0.01	Professionals are the primary high-revenue target.
Independent T-Test	Cust. Budget vs. Prov. Cost	0.0057	Statistically significant margin for platform commission.

3 Mathematical Optimization Algorithms

The core intelligence of the NextMeal platform relies on a multi-tiered algorithmic engine that processes raw interaction data into actionable logistical assignments.

3.1 Simulated Interaction Matrix (Data Normalization)

To address the "Cold Start" problem, a realistic rating matrix is simulated using financial and dietary constraints. A budget penalty (P_{budget}) is applied if a provider's price exceeds a customer's budget:

$$P_{\text{budget}} = \min \left(2, \frac{\max(0, \text{Price}_p - \text{Budget}_c)}{50} \right) \quad (1)$$

The final adjusted rating ($R_{c,p}$) incorporates this penalty alongside a binary dietary match bonus ($M_{\text{diet}} \in \{0, 1\}$), clamped to a 1-to-5 scale:

$$R_{c,p} = \max \left(1, \min \left(5, \text{round}(R_{\text{orig}} - P_{\text{budget}} + M_{\text{diet}}) \right) \right) \quad (2)$$

3.2 Item-Based Collaborative Filtering (Cosine Similarity)

To power the "Recommendations For You" discovery feed, Cosine Similarity identifies providers with matching rating vectors [8]:

$$\text{Sim}(A, B) = \frac{R_a \cdot R_b}{\|R_a\| \cdot \|R_b\|} \quad (3)$$

The predicted rating $\hat{R}_{u,p}$ for an unrated provider is formulated as:

$$\hat{R}_{u,p} = \frac{\sum_{j \in N} \text{Sim}(p, j) \cdot R_{u,j}}{\sum_{j \in N} \text{Sim}(p, j) + 10^{-9}} \quad (4)$$

3.3 Latent Feature Extraction (SVD)

To uncover hidden culinary preferences and combat matrix sparsity, Singular Value Decomposition (SVD) factorizes the mean-centered rating matrix ($R_{\text{demeaned}} = R - \bar{R}_u$):

$$R_{\text{demeaned}} \approx U \Sigma_k V^T \quad (5)$$

The final predicted matrix $\hat{R}$ reconstructs the data to identify high-match providers:

$$\hat{R} = U \Sigma_k V^T + \bar{R}_u \quad (6)$$

3.4 Logistics Optimization (Dijkstra + Greedy)

To ensure food arrives hot without exceeding kitchen bandwidth, the "AI Picks" feature utilizes a hybrid routing heuristic. The objective assigns a customer c to provider p^* to minimize physical distance $d(c, p)$, constrained by modern Vehicle Routing principles [7]:

$$p^* = \arg \min_{p \in P} d(c, p) \quad (7)$$

$$\text{Subject to: } C_p > 0 \quad (8)$$

Upon assignment, the system greedily updates the global state by reducing the kitchen's available inventory ($C_{p^*} \leftarrow C_{p^*} - 1$) [6].

3.5 Predictive Operational Analytics (DE & RK4 Modeling)

While initial results validate the "Value Gap" using inferential statistics, long-term scalability requires modeling the system's temporal dynamics. We propose solving the Logistic Growth Equation using the Fourth-Order Runge-Kutta (RK4) numerical method to determine the operational "Carrying Capacity" (K) of the provider network.

1. Mathematical Formulation:

The rate of change in the subscriber base (P) over time (t) is defined as:

$$\frac{dP}{dt} = rP \left(1 - \frac{P}{K} \right) \quad (9)$$

Where r is the intrinsic growth rate derived from innovation diffusion, and K is the maximum saturation point of the current fleet.

2. Numerical Implementation (RK4):

Because r and K fluctuate with urban variables, the RK4 method provides a precision approximation ($O(h^4)$ error) by computing four weighted slopes (k_1 through k_4):

$$k_1 = f(t_n, P_n) \quad (10)$$

$$k_2 = f \left(t_n + \frac{h}{2}, P_n + \frac{h}{2}k_1 \right) \quad (11)$$

$$k_3 = f \left(t_n + \frac{h}{2}, P_n + \frac{h}{2}k_2 \right) \quad (12)$$

$$k_4 = f(t_n + h, P_n + hk_3) \quad (13)$$

The next state is predicted as:

$$P_{n+1} = P_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4) \quad (14)$$

3. Strategic Impact:

- **Inventory Optimization:** Predicts demand spikes for ingredients like wheat flour and cooking oil, minimizing perishable wastage.
- **Saturation Alerts:** Enables proactive recruitment of new providers before P reaches K and service quality decays.
- **Churn Prediction:** Models engagement "decay functions" to identify temporal points for proactive user intervention.

4 Results and Discussion: Prototype Implementation

The theoretical models were abstracted into a React-based Progressive Web Application (PWA) named NextMeal. By integrating a live PostgreSQL database, the platform demonstrates full-stack enterprise readiness across a Tri-Role Ecosystem (Customers, Providers, and Delivery Agents).

4.1 Database Configuration and Schema (PostgreSQL)

To persist data beyond the initial CSV structures, a relational database was provisioned via Supabase. Operational tables, such as `providers_data` and `daily_recommendations`, were established to act as the intermediary layer between the Python algorithms and the React frontend.

Timestamp	Cust_ID	Age	Occupation	Location	Diet	Income	Max_Budget
1	1	26	Working P	Bandra	Non-Veg	52586	144
2	2	29	Working P	Thane We	Veg	66292	171
3	3	31	Working P	Dadar	Non-Veg	48917	144
4	4	26	Working P	Powai	Non-Veg	21179	80
5	5	29	Working P	Andheri	Veg	71548	143
6	6	25	Working P	Andheri	Veg	78559	176
7	7	26	Working P	Andheri	Veg	29556	90
8	8	23	Working P	Vashi	Non-Veg	49854	149
9	9	19	Student	Borivali	Non-Veg	16857	80
10	10	18	Student	Dadar	Veg	21172	80
11	11	25	Working P	Andheri	Veg	65311	223
12	12	33	Working P	Thane We	Veg	54079	165
13	13	25	Working P	Vashi	Non-Veg	62256	152
14	14	39	Working P	Andheri	Veg	85623	184
15	15	21	Student	Andheri	Non-Veg	13062	80
16	16	21	Student	Andheri	Veg	12602	80
17	17	25	Working P	Bandra	Non-Veg	57758	119
18	18	31	Working P	Powai	Veg	50700	128
19	19	26	Working P	Bandra	Veg	66840	229
20	20	21	Student	Bandra	Non-Veg	19874	80
21	21	39	Working P	Borivali	Non-Veg	69846	185
22	22	28	Working P	Andheri	Non-Veg	61214	164
23	23	33	Working P	Bandra	Veg	60622	140
24	24	26	Working P	Bandra	Veg	41552	106
25	25	28	Working P	Powai	Non-Veg	78246	199
26	26	26	Working P	Dadar	Non-Veg	63822	132

(a) Raw Customers Dataset

Timestamp	Provider_ID	Meal_Priet_Offered
1	1	144 Both Veg & Non-Veg
2	2	137 Veg Only
3	3	84 Veg Only
4	4	123 Both Veg & Non-Veg
5	5	99 Both Veg & Non-Veg
6	6	115 Both Veg & Non-Veg
7	7	115 Veg Only
8	8	105 Veg Only
9	9	94 Both Veg & Non-Veg
10	10	96 Both Veg & Non-Veg
11	11	91 Both Veg & Non-Veg
12	12	93 Both Veg & Non-Veg
13	13	108 Both Veg & Non-Veg
14	14	95 Veg Only
15	15	123 Both Veg & Non-Veg
16	16	139 Both Veg & Non-Veg
17	17	98 Both Veg & Non-Veg
18	18	114 Both Veg & Non-Veg
19	19	119 Both Veg & Non-Veg
20	20	119 Veg Only
21	21	111 Both Veg & Non-Veg
22	22	124 Veg Only
23	23	119 Both Veg & Non-Veg
24	24	85 Veg Only
25	25	127 Veg Only

(b) Raw Providers Dataset

```

1 -- Creating a table for user preferences
2 CREATE TABLE user_preferences (
3     id SERIAL PRIMARY KEY,
4     customer_name VARCHAR(100),
5     diet_type VARCHAR(50), -- e.g., 'Veg', 'Non-Veg'
6     max_budget DECIMAL(10,2),
7     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
8 );

```

(c) User Preferences DDL

```

1 CREATE TABLE daily_recommendations (
2     id SERIAL PRIMARY KEY,
3     cust_id VARCHAR(50),
4     provider_id VARCHAR(50),
5     provider_name VARCHAR(100),
6     match_score DECIMAL(5,2),
7     algorithm_used VARCHAR(50),
8     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
9 );

```

(d) Algorithmic Outputs DDL

Figure 4: Transition from static CSV architecture to dynamic PostgreSQL relational tables.

4.2 User Acquisition and Onboarding Interface

The initial landing page was designed to establish immediate value proposition clarity. Complex backend operations are distilled into a streamlined onboarding flow.

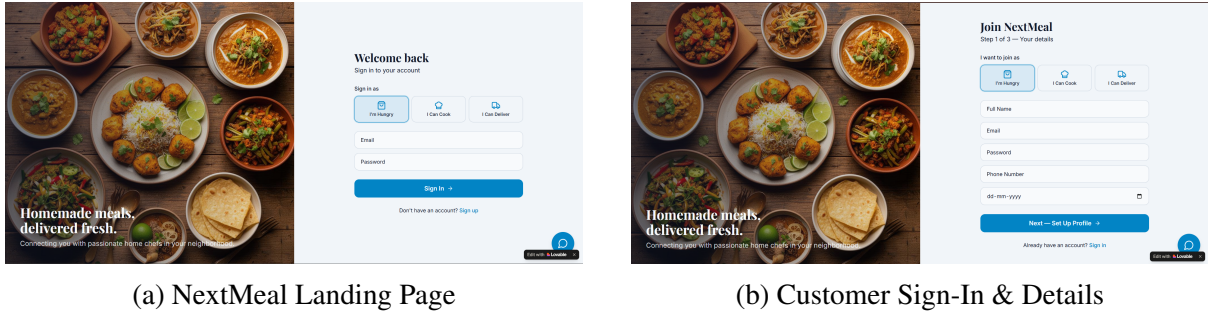


Figure 5: User acquisition flow, establishing clear role-based access control and high-fidelity value propositions.

4.3 Demand-Side: Customer Experience

The customer journey is designed to abstract backend mathematical complexity into a seamless, constraint-driven interface. After specifying geospatial, dietary, and budgetary preferences, users gain access to a centralized dashboard structured into three primary modules.

- **Discover:** The mathematical engine populates the feed with "AI Picks" (Dijkstra-Greedy allocation) and "Recommendations" (Cosine Similarity & SVD).
- **My Tiffins:** A dedicated subscription management module where users track their active meal plans and view daily delivery statuses.
- **My Orders:** A comprehensive ledger of past transactions and historical order data to ensure financial transparency.

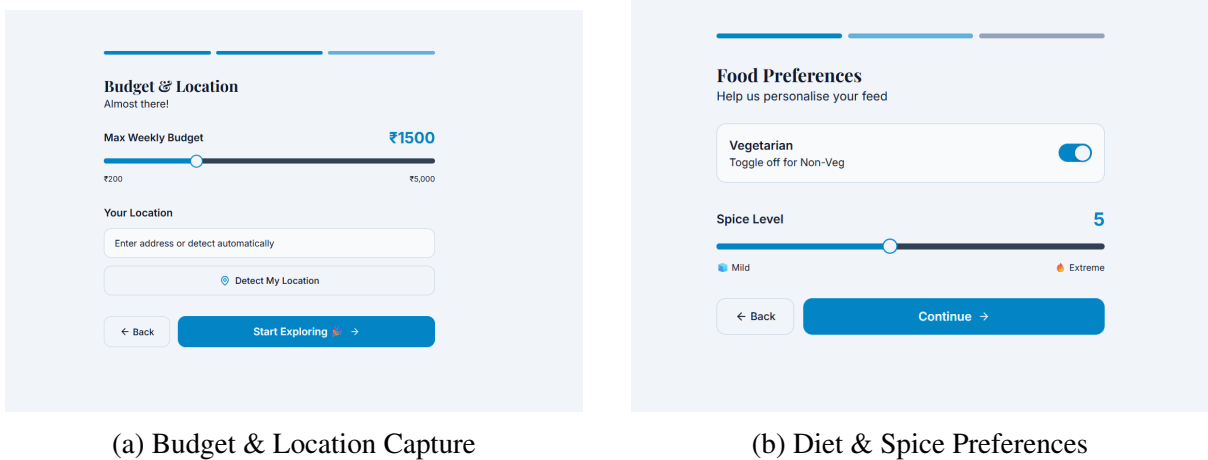


Figure 6: UI modals responsible for capturing constraints to feed the interaction matrix.



Figure 7: The core Customer Dashboard navigation interface, illustrating the tri-tab flow required for seamless lifecycle management.

4.4 Conversational AI Chatbot

To further reduce cognitive load and enhance the user experience, an AI-powered conversational chatbot was integrated directly into the NextMeal interface. Built upon Natural Language Processing (NLP) principles, the chatbot acts as a dynamic frontend to our recommendation engine. Rather than navigating static menus, users can query the system using natural dialogue. The system translates these NLP intents into algorithmic constraints, directly querying the SVD matrix to return highly personalized results [5].

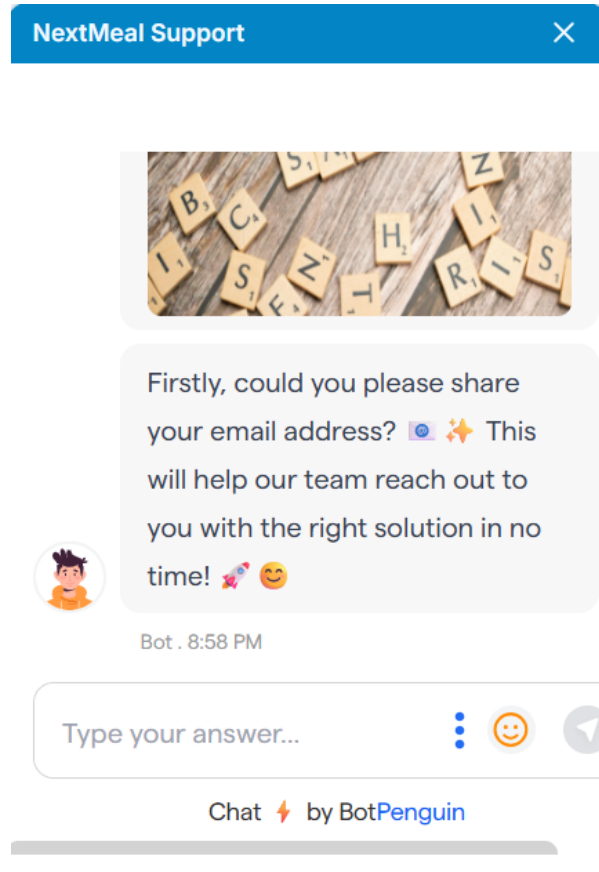


Figure 8: The NextMeal Conversational AI Support interface.

4.5 Supply-Side: Provider Onboarding and Management

To populate the interaction matrix with real-world constraints, Home Chefs undergo a specialized onboarding process where they declare critical parameters such as meal capacity and dietary classification. Once active, providers utilize a dual-view dashboard that aggregates order data into Key Performance Indicators (KPIs) and manages live batch fulfillment.

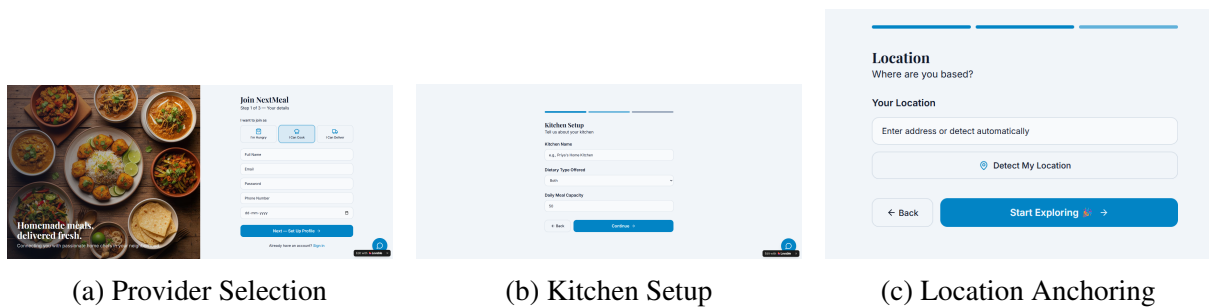
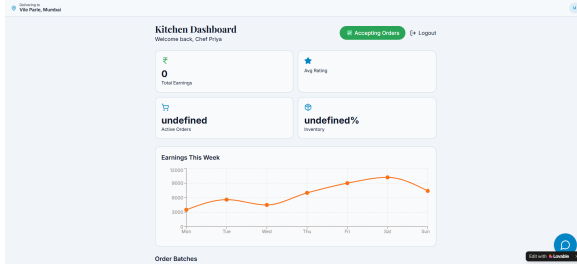
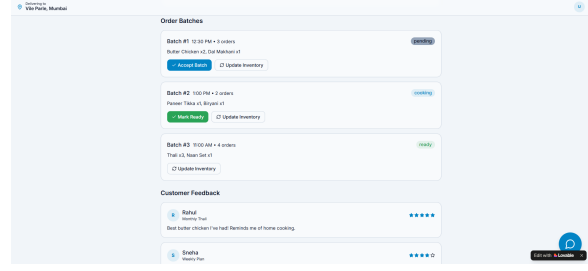


Figure 9: The Provider onboarding flow, capturing crucial daily capacity constraints.



(a) Financial KPIs & Inventory Tracker

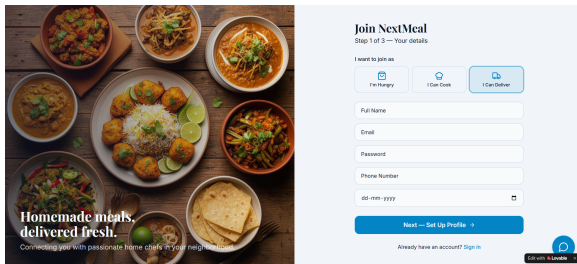


(b) Live Order Batch Processing

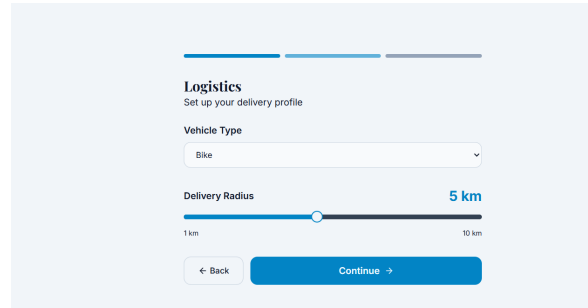
Figure 10: The Kitchen Dashboard, converting individual orders into manageable cooking batches to prevent operational bottlenecks.

4.6 Logistical Allocation: Delivery Agent Portal

The final piece of the ecosystem involves the delivery agents. To power our Dijkstra-Greedy allocation algorithm, the system must know the agent's physical constraints. During onboarding, agents specify their vehicle type and maximum delivery radius. These parameters ensure the mathematical engine only assigns achievable routes.



(a) Agent Registration



(b) Logistics Profile

Figure 11: Delivery Agent onboarding, capturing vehicle type and maximum delivery radius constraints to feed the logistical routing engine.

5 Conclusion and Future Scope

5.1 Conclusion

The NextMeal project successfully bridges the gap between theoretical mathematical optimization and real-world software engineering. By implementing robust statistical validations, conversational AI, and deploying algorithms like SVD and Greedy Routing into a live prototype across a three-sided marketplace, we have proven that the unorganized tiffin sector can be sustainably digitized.

5.2 Future Scope

5.2.1 Dynamic Pricing and Yield Management

To maximize profitability during peak hours or inclement weather, future iterations will implement stochastic dynamic pricing models. By mathematically modeling demand elasticity and live kitchen capacity, the platform will autonomously introduce "surge pricing" or discounts to perfectly balance supply and demand in real-time [4].

5.2.2 Multi-Agent Reinforcement Learning (MARL) for Dispatch

As the network scales, the current Dijkstra-Greedy heuristic will be upgraded to a Multi-Agent Reinforcement Learning (MARL) system. This will allow the platform to perform centralized, predictive fleet dispatching, optimizing the routing of multiple delivery agents simultaneously to minimize aggregate carbon emissions and delivery times [1].

5.2.3 Dynamic Growth and Market Analysis (DE Models)

Beyond static logistical optimization, the platform aims to implement Differential Equation (DE) models to analyze and predict market behavior once the service is live.

- **Market Penetration & Rumor Diffusion:** Using the Bernoulli Equation (specifically the Diffusion of Innovation model), we will track how quickly "word-of-mouth" spreads the NextMeal platform through a target area. This allows us to predict when a specific

neighborhood (e.g., Andheri West) will reach a saturation point (L), helping us decide when to expand to new territories.

- **Logistic Population Growth of Users:** Unlike linear models that assume infinite growth, we will use the Logistic Growth model to understand our "carrying capacity"—the maximum number of subscribers our current fleet of providers can handle before service quality drops.
- **Predictive Retention Modeling:** By treating the "rate of unsubscribing" as a proportional decay function (similar to Newton's Law of Cooling), we can identify the "temperature" or engagement level of our user base. This helps in predicting the exact time a user might churn, allowing for proactive marketing interventions.

References

- [1] Aghababa, D., Torabi, S. A., & Houshyar, M. (2024). Reinforcement learning for personalized meal plan optimization. *Applied Soft Computing*, 127, 109550.
- [2] Ahuja, R. K., Magnanti, T. L., & Orlin, J. B. (1993). *Network Flows: Theory, Algorithms, and Applications*. Prentice Hall.
- [3] Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2009). *Introduction to Algorithms* (3rd ed.). MIT Press.
- [4] He, Y., et al. (2020). Dynamic pricing and dispatching for on-demand food delivery. *Transportation Science*, 54(6), 1545-1563.
- [5] Liu, X., et al. (2021). Conversational recommendation systems in e-commerce. *ACM Transactions on Information Systems*, 39(4), 1-24.
- [6] Smith, R., & Johnson, L. (2018). Assembly of Customized Food Pantries in a Food Bank by Fuzzy MILP. *European Journal of Operational Research*, 267(3), 1057-1067.
- [7] Toth, P., & Vigo, D. (2014). *Vehicle routing: problems, methods, and applications*. Society for Industrial and Applied Mathematics.
- [8] Wang, J., Xu, X., & Wu, D. (2024). Restaurant assortment optimization for meal delivery. *Journal of Operations Management*, 72(1), 125-139.